Last time we were studying interference from an aperture with two slits and we found:

constructive interference: $y_m = m\left(\frac{R\lambda}{d}\right)$ $m = 0, \pm 1, \pm 2, \ldots$

destructive interference: $y_m = \left(m+\frac{1}{2}\right)\left(\frac{R\lambda}{d}\right)$ $m = 0, \pm 1, \pm 2, \ldots$

where y is the height above ($y>0$) or below ($y<0$) the midline on the screen, level with the two slits. This leads to a series of bright fringes and dark bands, as can be seen on the accompanying slide.

Example: light from a HeNe laser ($\lambda = 632.8$ nm) is incident on 2 slits separated by 0.125 mm. What is the distance between the $m=-2$ and $m=2$ bright fringes on a screen 3.70 m away?

Solution: This is constructive interference, so use $y_m = \frac{mR\lambda}{d}$

For $m=-2$: $y_{-2} = (-2)\left(\frac{(3.70\,\text{m})(632.8\times10^{-9}\,\text{m})}{0.125\times10^{-3}\,\text{m}}\right) = -37.5\,\text{mm}$

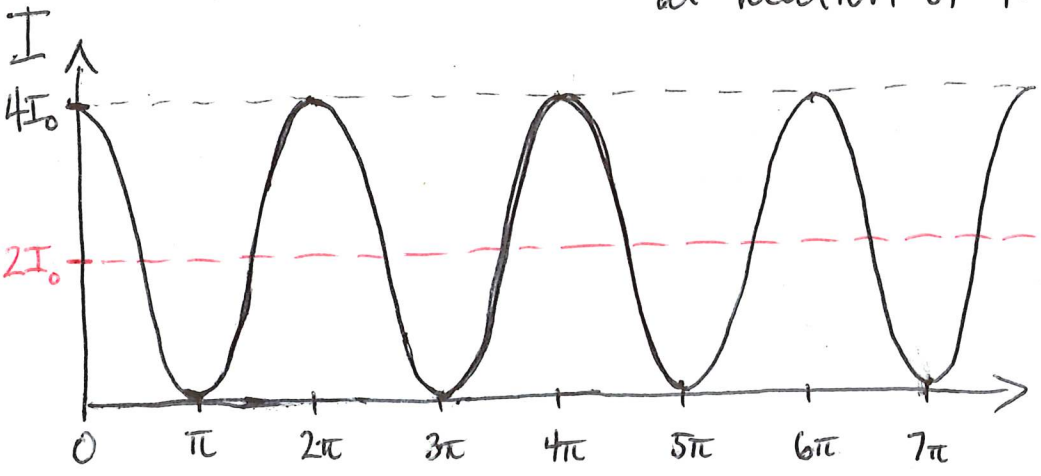
$m=+2$ will be at $y_2 = +37.5$ mm

$\Rightarrow$ $\boxed{\Delta y = 75.0\,\text{mm}}$

What does the intensity pattern look like? Recall that $I \propto A^2$ where A is the resultant wave amplitude. If each wave has an amplitude a, then the resultant wave maxima (constructive interference) has amplitude $2a$. Define I_0 as the intensity of incident light on the 2 slits so that $I_0 \propto a^2$, then we can see at the maxima $I = 4I_0$ while at the minima $I = 0$ (destructive interf.) The intensity varies according to

$$I = 4I_0 \cos^2\left(\tfrac{1}{2}\phi\right)$$

where $\phi = k\Delta r = \frac{2\pi}{\lambda} d\sin\theta$ is the phase difference b/w 2 waves at location of the screen.

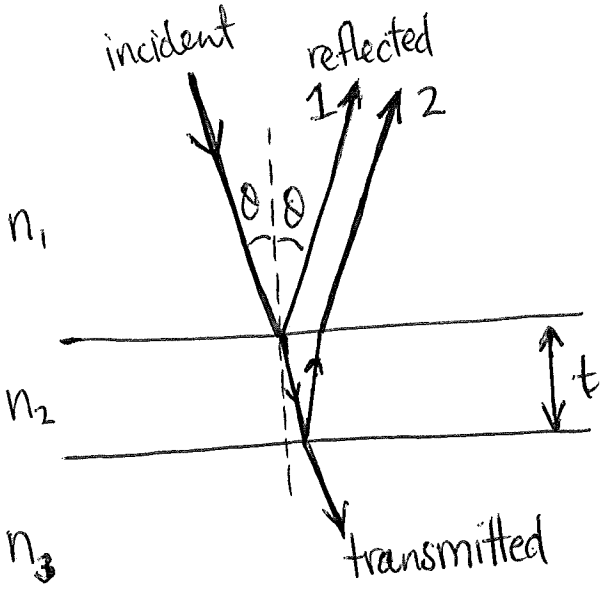


bright fringes are the maxima

dark bands are the minima

The average intensity is $\langle I \rangle = 4I_0 \langle \cos^2\left(\tfrac{1}{2}\phi\right) \rangle$ but $\langle \cos^2\theta \rangle$ over one period is $\frac{1}{2}$, so $\langle I \rangle = 2I_0$ which must be the case since each slit emits light with intensity I_0. The interference pattern redistributes the intensity.

Another important interference effect that will use many of the same concepts in a different context is <u>thin film interference</u>.



An incident ray hits a film of thickness t and is partly reflected and partly transmitted. The transmitted wave hits the next boundary and is partly reflected/transmitted by that surface too. The two reflected waves can then interfere.

We will assume that the angle of incidence is small so that the extra path length of ray 2 is simply $2t$. The two rays will constructively interfere if their phases differ by

$$\Delta\phi = 2\pi m \quad m = 0, 1, 2, \ldots$$

and destructively interfere if their phases differ by

$$\Delta\phi = 2\pi\left(m + \tfrac{1}{2}\right) \quad m = 0, 1, 2, \ldots$$

We need to consider 2 things when evaluating $\Delta\phi$:

1) path length difference b/w the two waves (same idea as for two slit interference)
2) π phase shifts from the reflections.

5 minute break

For acoustic waves we introduced the impedance $Z = \frac{\#}{V}$ and we found that when a wave reflects off a boundary in which $Z_1 > Z_2$ then $R > 0$ and the reflected wave is in phase with the incident wave (no phase shift), but when $Z_2 > Z_1$, then $R < 0$ and the reflected wave is out of phase with the incident wave (π phase shift).

For light we define the <u>index of refraction</u>

$$n = \frac{c}{v}$$

c = speed of light in vacuum
v = speed of light in the medium

which plays the role of an <u>optical impedance</u>. When light reflects off a surface:

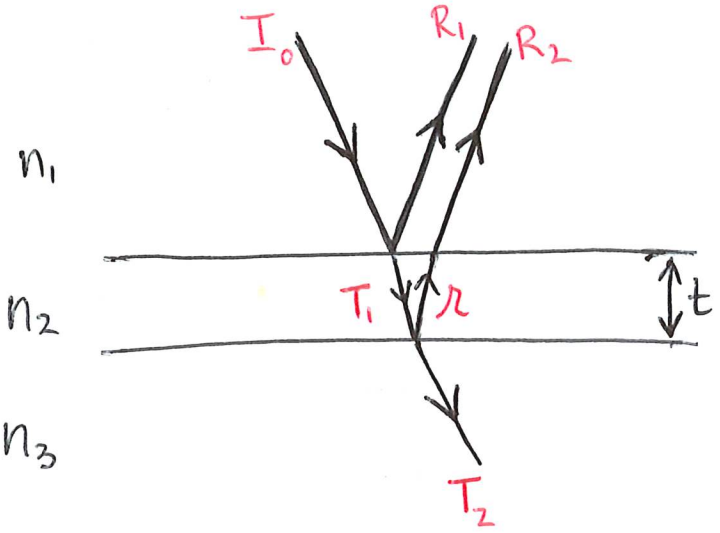
if $n_1 > n_2 \Rightarrow$ no phase shift
if $n_2 > n_1 \Rightarrow$ π phase shift

This is because the reflection and transmission coefficients are given by

$$R = \frac{n_1 - n_2}{n_1 + n_2} \quad \text{and} \quad T = \frac{2n_1}{n_1 + n_2}$$

(just replace Z with n)

Let's now look at this in more detail to see how to analyze thin film interference.



Aside: if $n_1 = 1$ for air and $n_2 = 1.33$ for water, then $R \approx 0.02$ and $T \approx 0.98$ so most of the light is transmitted. If incident ray has intensity I_0:

- R_1 has $0.02 I_0$ and T_1 has $0.98 I_0$
- r has $0.0196 I_0$ and T_2 has $0.96 I_0$
- R_2 has $0.0192 I_0$
- Higher order reflections and transmissions carry diminishing intensities so we can ignore them.

The conditions for interference b/w R_1 and R_2 in terms of the path length difference is:

constructive: $r_2 - r_1 \approx \boxed{2t = m\lambda + \epsilon\frac{\lambda}{2}}$ $m = 1, 2, \ldots$ $\epsilon = 0, 1, \text{or } 2$

↑ from π phase shifts on reflection

destructive: $r_2 - r_1 \approx \boxed{2t = \left(m - \frac{1}{2}\right)\lambda + \epsilon\frac{\lambda}{2}}$ $m = 1, 2, \ldots$ $\epsilon = 0, 1, \text{or } 2$

Important: the frequency of light does not change when going from one material to another, just like other waves, but the speed does and hence so does the wavelength λ:

$$f\lambda = v \quad \text{where} \quad v = \frac{c}{n} \Rightarrow \lambda = \frac{c}{n}\frac{1}{f} \quad \text{but} \quad \lambda_0 = \frac{c}{f}$$

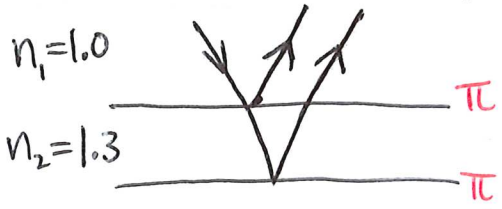
where λ_0 is the wavelength in vacuum (or air since $n_{air} \approx 1$)

$$\Rightarrow \boxed{\lambda = \frac{\lambda_0}{n}}$$ wavelength in medium

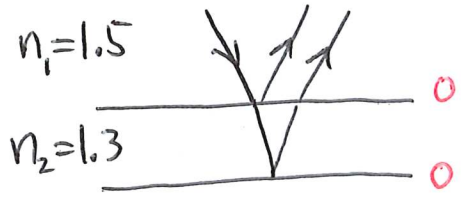
$\Rightarrow$ constructive: $\boxed{2t = \frac{m\lambda_0}{n} + \frac{\epsilon\lambda_0}{2n}}$

destructive: $\boxed{2t = \left(m-\frac{1}{2}\right)\frac{\lambda_0}{n} + \frac{\epsilon\lambda_0}{2n}}$

Case 1: no relative phase shift ($\epsilon = 0$ or 2)



or



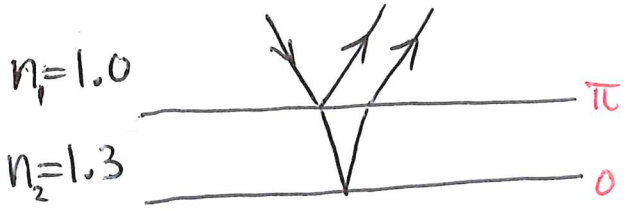
both reflections have π phase shift $\Rightarrow \epsilon = 2$

both reflections have no phase shift $\Rightarrow \epsilon = 0$

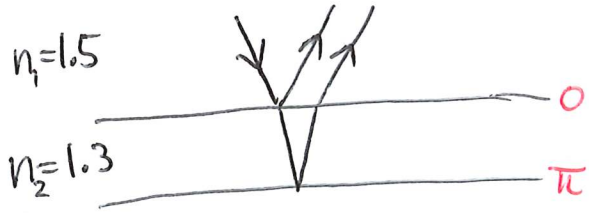
constructive: $\boxed{2t = m\frac{\lambda_0}{n}}$ $m = 1, 2, \ldots$

destructive: $\boxed{2t = \left(m-\frac{1}{2}\right)\frac{\lambda_0}{n}}$ $m = 1, 2, \ldots$

Case 2: π phase shift ($\epsilon = 1$)



or



constructive: $\boxed{2t = \left(m-\frac{1}{2}\right)\frac{\lambda_0}{n}}$ $m = 1, 2, \ldots$

destructive: $\boxed{2t = m\frac{\lambda_0}{n}}$ $m = 1, 2, \ldots$